

Unghui Nam

Graduate Student, Seoul National University

Research Interests: Vision-Language-based Manipulation, Robotic Data Representation

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About Me

I am a graduate student working on **Vision-language-based Manipulation** and **Robotic Data Representation**, advised by Prof. Byonghyo Shim at Seoul National University. My research focuses on building robot learning system that imitates the human intelligence, with the goal of designing autonomous robotic systems that can plan and execute actions based on vision and language perceptions.

Education

Seoul National University *Mar. 2020 – Feb. 2026*
B.S. in Mechanical and Aerospace Engineering, School of Engineering
(*cum laude*)

Seoul National University *Mar. 2026 – Present*
M.S.&Ph.D. in Electrical and Computer Engineering, School of Engineering

Publications

- [1] **Adaptive Capacity Allocation for Vision Language Action Fine-tuning**
D.H. Kim, M.J. Bae*, **U. Nam***, G.H. Kim*, S.Y. Lee*, K.H. Shim, B. Shim (*Equal contribution.)
IEEE International Conference on Robotics and Automation (ICRA), 2026.
- [2] **Learning Decomposed Visibility for Efficient Active Exploration of Cluttered Scenes**
S.Y. Lee*, M.S. Choi*, J.H. Gong, **U. Nam**, M.J. Bae B. Shim
Conference on Robot Learning (CoRL), 2026.
- [3] **Scalable Autonomous Robotic Data Collection via Data Affordance-Guided Intervention**
U. Nam, M.J. Bae, S. Kim, J.A. Kim, S.Y. Lee, M.S. Choi, B. Shim
Under Review.
- [4] **DexStitch: Learning and Composing Dexterous Skills from Human Videos via Contact-Aware Robotic Inpainting**
B. Choi*, S. Jung*, **U. Nam***, Y. Song*, S. Min, M. Lee, B. Shim, Y. Lee
Under Review.

Academic Service

- Conference Reviewer, Conference on Robot Learning (CoRL), 2026

Industry Experience

Whatssub Inc. *Jan. 2023 – Jun. 2023*
Data Analyst

- Designed payment-transaction categorization schemes and built an automated classification pipeline.
- Developed an LLM-based system that generates personalized marketing copy for target user

segments, and integrated it into an automated production workflow.

Selected Projects

Piuda Project – SW Development Competition

May 2024 – Jul. 2024

Project Lead & Designer, App for Supporting Homeless Individuals

- Led overall application planning and UI/UX design.
- Advanced to the finals of the competition.

Multinational ALPR in Cameras with Low Resolution and Low Frame Rate

Sep. 2024 – Dec. 2024

Course Project, Fundamentals of Deep Learning, SNU

- Surveyed and reimplemented existing license-plate detection and recognition models.
- Led benchmark evaluation on the dataset and authored the final technical report.

Automated Generation of Paint-by-Number Templates

Sep. 2024 – Dec. 2024

Course Project, Computer Vision, SNU

- Designed and implemented edge-detection, blurring, and color-clustering modules for an automated paint-by-number template generator.
- Conducted performance evaluation of the end-to-end pipeline.

Zumo Shield Robot

Sep. 2024 – Dec. 2024

Course Project, Aerospace Sensor Systems, SNU

- Implemented line-tracing and object-avoidance behaviors using infrared and ultrasonic sensors.
- Developed a self-localization module fusing gyroscope and accelerometer measurements.

Automating Robotic Manipulation Data Collection

Dec. 2025 – Feb. 2026

Project Lead

- Implemented a fully automated robotic manipulation data collection pipeline.
- Integrated VLM agents and segmentation modules into 3D object point cloud detector.

Technical Skills

Programming	Python, C++, Linux
Robotics	ROS 2
Languages	Korean (Native), English (Fluent)

Certifications & Language Tests

TOEFL iBT	Score: 104	Sep. 2016
Craftsman Information Processing	National Technical Qualification	Apr. 2023
TEPS	Score: 497	Aug. 2025